Haoyang Zhang

Electricity Markets | Optimization | Flexibility

Belgium | zhanghaoyang1993.github.io | LinkedIn | GitHub

Professional profile

Electricity market modeller with a PhD from Eindhoven University of Technology. Develops quantitative models for European market clearing, market design and demand-side flexibility, combining optimization, public market data and multi-agent simulation.

Experience

Energy Market Modelling Expert | VITO / EnergyVille
November 2025 - present · Belgium

CROCODILE and EU-DREAM: European market reconstruction and residential flexibility modelling.

Visiting Researcher | Empa
March - July 2025 · Switzerland

- Investigated bidding across sequential local markets using hierarchical multi-agent reinforcement learning; co-authored a research preprint with Empa and TU/e collaborators.

Doctoral Researcher | Eindhoven University of Technology
October 2021 - October 2025; PhD awarded June 2026 · Netherlands

- Developed a simulation framework for local electricity and flexibility markets to study strategic bidding under distribution network constraints.
- Compared learned bidding behaviour with truthful-bidding benchmarks and assessed multi-agent learning reliability using a no-regret measure.
- Evaluated pricing mechanisms and market power detection methods in simulated local markets.

Associate Researcher | China Southern Power Grid Energy Development Research Institute
2019 - 2021 · China

Energy policy and electricity market analysis.

Current projects

CROCODILE | Offshore Bidding Zones & Market Coupling

- Develop an EUPHEMIA-style reconstruction of Core day-ahead market clearing using public JAO capacity data and NEMO market bid information.
- Use historical reconstruction to support assessment of offshore bidding-zone configurations and European market-integration arrangements.

EU-DREAM | Residential Flexibility

- Develop optimization-based household flexibility models covering heat pumps and EV charging under dynamic electricity prices and capacity-based network tariffs.

Selected research & personal project

MEGAMIND

Eindhoven University of Technology | PhD research completed in 2026

- Designed a hierarchical local market framework using multi-cut Benders decomposition; published the method in Applied Energy (2024).
- Assessed multi-agent reinforcement learning reliability for hybrid local market simulation; published in Applied Energy (2025).

EuroPowerQuantLab

Independent personal project | Under development

- Developing a Python research platform for European power-market data, forecasting research and historical backtesting; capabilities remain under development.

Education

PhD in Electrical Engineering | Eindhoven University of Technology | 2026

MSc in Electric Power Engineering | KTH Royal Institute of Technology | 2019
Exchange & master's thesis research at ETH Zürich, Power Systems Laboratory (2017 - 2018); distribution grid state estimation.

BSc in Electrical Engineering and Automation | South China University of Technology | 2016

Technical skills

Electricity Markets & Power Systems: EUPHEMIA-style clearing; Core FBMC; published CNEC/PTDF/RAM parameters; offshore bidding zones; flexibility markets; congestion management

Optimization & Operations Research: Mathematical optimization; convex/non-convex models; decomposition; multi-cut Benders; heuristic/evolutionary algorithms

AI & Quantitative Methods: Reinforcement learning; MARL; deep reinforcement learning; agent-based simulation; quantitative market analysis

Programming & Modelling: Python; optimization model development; market simulation; historical data analysis; validation and reproducible workflows

Selected publications

Haoyang Zhang, Sen Zhan, Koen Kok, Nikolaos G. Paterakis. **Establishing a hierarchical local market structure using multi-cut Benders decomposition**. *Applied Energy*, 2024. DOI

Haoyang Zhang, Dawei Qiu, Koen Kok, Nikolaos G. Paterakis. **Reliability assessment of multi-agent reinforcement learning algorithms for hybrid local electricity market simulation**. *Applied Energy*, 2025. DOI

Xin Li, Sen Zhan, **Haoyang Zhang**, Koen Kok, Nikolaos G. Paterakis. **A privacy-preserving distributed congestion management framework via implicit differentiation bilevel optimization**. *Electric Power Systems Research*, 2027 (online 2026; volume 2027). DOI

Awards & fellowships

2025: EEM Best Student Paper Award | European Energy Market conference

2025: NCCR Automation Fellowship | NCCR Automation / Empa research visit

Professional training & certifications

November 2024: The Physical and Financial Power Markets Course | Nord Pool Academy

October 2021: Fundamentals of Reinforcement Learning | University of Alberta